

TASVIRLARNI TAHLIL QILISH VA ANGLASH · Amaliy mashg'ulot (Practice)

10-amaliy mashg'ulot. Semantik segmentatsiya

2026–2027 o'quv yili

Mavzu. Pikel darajasida sinflash: FCN, U-Net, DeepLab; skip connection; transposed convolution; baholash (PA, IoU, mIoU, Dice); cross-entropy va Dice loss; klassik usul bilan taqqoslash.

Maqsadlar. Talaba: conv/transposed-conv o'lchamlari va receptive field ni qo'lda hisoblaydi; confusion matrix dan mIoU va Dice ni topadi; Dice va CE loss ni hisoblaydi va isbotlaydi; PyTorch da kichik U-Net ni yozadi, o'qitadi va ablyatsiya qiladi; natijani klassik Otsu bilan solishtiradi.

Muhit. Python 3.11 yoki 3.12, faqat CPU. `python -m venv .venv`, so'ng `pip install -r requirements.txt` (numpy 2.4.4, scipy 1.17.1, scikit-image 0.26.0, opencv-python-headless 4.13.0.92, torch 2.14.0, torchvision 0.29.0). Barcha tasodifiylik `seed=42`; kod boshida `torch.manual_seed(42)`, `torch.use_deterministic_algorithms(True)`, `torch.set_num_threads(1)`. Sonlar 4 xonagacha yaxlitlanadi.

Baholash (100 ball). A (1–13): har biri 2 ball; B (14–17): har biri 4 ball; C (18–30): 27 va 28-topshiriq 7 balldan, qolganlari 4 balldan. Kod topshiriqlari bitta `solutions.py` faylida, har topshiriq — alohida funksiya; natija so'ralgan formatda chop etiladi.

A. Qo'lda hisoblash

- Conv o'lchamlari.** $o = \lfloor (i + 2p - d(k - 1) - 1)/s \rfloor + 1$. Kirish 128×128 : $\text{Conv}(k = 3, p = 1, s = 1) \rightarrow \text{MaxPool}(2) \rightarrow \text{Conv}(k = 3, s = 2, p = 1) \rightarrow \text{Conv}(k = 5, p = 0)$. Har qadamdan keyingi tomonni toping.
- Transposed convolution.** $o = (i - 1)s - 2p + k + a$ bo'yicha: (a) $i = 16, k = 2, s = 2, p = 0$; (b) $i = 7, k = 4, s = 2, p = 1$; (c) $i = 10, k = 3, s = 2, p = 1, a = 1$; (d) $i = 5, k = 3, s = 1, p = 0$.
- Receptive field.** Qatlamlar: conv3, conv3, pool2, conv3 (dilation 2), pool2, conv3. $r_\ell = r_{\ell-1} + (k_{\text{eff}} - 1)j_{\ell-1}$, $j_\ell = j_{\ell-1}s_\ell$, $r_0 = j_0 = 1$. Har qatlamdan keyin (r, j) ni yozing.
- Atrous convolution.** $k = 3$ uchun $r = 6, 12, 18$ (ASPP) da k_{eff} ni toping. `Conv2d(256, 256, 3, dilation=6)` ning parametrlar soni nechta va u $r = 1$ dagidan farq qiladimi?
- Confusion matrix.** Qator — haqiqat, ustun — bashorat: $M = \begin{pmatrix} 120 & 6 & 4 \\ 8 & 40 & 2 \\ 2 & 3 & 15 \end{pmatrix}$. PA, IoU_c, mIoU, Dice_c, mDice ni toping.
- Ikkilik niqob.** 8×8 to'rtburchak haqiqiy obyekt T = qatorlar 2.4, ustunlar 2.5; bashorat P — xuddi shu to'rtburchak bir ustun o'ngga surilgan. TP, FP, FN , IoU, Dice? So'ng: Dice = 0,8 bo'lsa IoU nechaga teng? IoU = 0,5 bo'lsa Dice?
- “Hammasi fon” modeli va maydon.** Fon ulushi 0,90, sinflar $C = 4$. Hamma pikselni fon deydigan modelning PA va mIoU si? Sun'iy yo'ldosh tasvirida (10 m/piksel) ekin niqobi 12 500 piksel: maydon necha gektar?
- Siljish sezgirligi.** Tomoni $a = 10$ kvadrat niqob (i) 1 piksel gorizontal, (ii) 1 piksel gorizontal va 1 piksel vertikal siljidi. Ikki holatda IoU ni toping; $a = 50$ uchun (i) ni hisoblang.
- Parametrlar soni.** (a) `Conv2d(3, 64, 3)`; (b) `ConvTranspose2d(128, 64, 2, stride=2)`; (c) U-Net dekoderida 64 kanalli up-xarita 64 kanalli skip bilan ulanadi, keyingi `Conv2d(?, 64, 3)`; (d) 1×1 boshcha $16 \rightarrow 5$ sinf.
- Loss qo'lda.** 4 piksel, $t = (1, 1, 0, 0)$ (1 — obyekt), $p_{\text{obj}} = (0,8; 0,6; 0,3; 0,1)$. CE, Dice loss ($\varepsilon = 0$, 2 sinf bo'yicha o'rtacha) va $w = (0,5; 1,5)$ bilan weighted CE (PyTorch ta'rifi: $\sum w_t \ell / \sum w_t$).
- Sinf og'irliklari.** Ulushlar (0,80; 0,15; 0,05). $w_c \propto 1/\sqrt{n_c}$, $\sum w_c = 3$ bo'yicha w ni toping.
- 1D transposed convolution.** $y = (1, 2)$, $w = (1, 2, 1)$, $s = 2$, $p = 0$. $z = C^T y$ ni yozing (C — mos konvolyutsiya matritsasi) va har pozitsiya nechta hissa olishini ko'rsating.
- Bilinear.** $f_{00} = 2, f_{10} = 6, f_{01} = 4, f_{11} = 10$. $f(0,25; 0,75)$ ni toping. Niqobni kattalashtirishda nega nearest ishlatiladi?

B. Keltirib chiqarish va tushunchalar

- Dice = $2J/(1 + J)$ ni isbotlang, $J \leq \text{Dice}$ ekanini ko'rsating. mIoU va mDice ikki modelni turlicha tartiblaydigan 2-sinfl misol tuzing.
- $o = (i - 1)s - 2p + k$ ni konvolyutsiya matritsasining transpozitsiyasi sifatida keltirib chiqaring; $s \mid k$ da har ichki pozitsiya aynan k/s hissa olishini, $s \nmid k$ da checkerboard hosil bo'lishini isbotlang.
- Soft Dice loss ning $\partial/\partial p_c(u)$ hosilasini toping. Nega kichik sinfda gradient kattaroq? Nega CE “hammasi fon” modelini yetarli jazolamaydi?
- Qisqa javob: (a) semantic, instance, panoptic segmentatsiya farqi; Mask R-CNN niqob boshchasi va RoIAlign; (b) FCN, U-Net, DeepLab (ASPP) asosiy g'oyalari; (c) bo'sh sinf ($T_c = P_c = \emptyset$) IoU si qanday olinadi; (d) qaysi augmentatsiya niqobga ham qo'llanadi?

C. Dasturlash

Umumiy maʼlumot (24–30-topshiriqlar): sunʼiy sahnalar (64×64 , sinflar: 0 — fon, 1 — toʻrtburchak, 2 — doira; yoritish qiyaligi va shovqin). **Aynan** quyidagi generator va tartib ishlatiladi:

```
def make_scenes(n, rng):
    X = np.zeros((n, 1, 64, 64), np.float32); Y = np.zeros((n, 64, 64), np.int64)
    yy, xx = np.mgrid[:64, :64]
    for i in range(n):
        g = rng.uniform(-1, 1, 2); img = 60 + 40*(g[0]*xx + g[1]*yy)/64
        x0, y0 = rng.integers(2, 30, 2); w, h = rng.integers(10, 22, 2)
        Y[i, y0:y0+h, x0:x0+w] = 1
        cx, cy = rng.integers(12, 52, 2); r = rng.integers(5, 10)
        Y[i][(xx-cx)**2 + (yy-cy)**2 <= r*r] = 2
        img = img + 70*(Y[i] == 1) + 120*(Y[i] == 2) + rng.normal(0, 12, (64, 64))
        X[i, 0] = img/255
    return X, Y
rng = np.random.default_rng(42)
train, val, test = make_scenes(200, rng), make_scenes(40, rng), make_scenes(100, rng)
```

18. `conf_mat(gt, pred, C)` (`np.bincount` orqali) va `seg_metrics(M)` → `dict(PA, IoU, mIoU, Dice, mDice)` yozing; boʻsh sinf `nan` boʻlsin, oʻrtacha `np.nanmean`. 5-topshiriq matritsasi va 6-topshiriq niqoblarida tekshiring; $\max |Dice - 2J/(1 + J)|$ ni chiqaring.
19. `out_conv(i,k,s,p,d)`, `out_tconv(i,k,s,p,a)` funksiyalari. 1 va 2-topshiriq konfiguratsiyalari uchun natijani `nn.Conv2d/nn.ConvTranspose2d` chiqish shakli bilan solishtiring (`torch.zeros` kirish). Format: `formula=.. torch=..`
20. `receptive_field(layers)`, `layers=[(k,s,d),...]` → `[(r,j),...]`. 3-topshiriq tarmogʻini PyTorch da quring (`conv` ogʻirliklari 1, `bias=False`, `pool` oʻrniga `AvgPool2d(2)`), 64×64 kirishda markaziy chiqish elementi gradientining nolmas sohasi tomonini oʻlchang.
21. Checkerboard: `ConvTranspose2d(1,1,k, stride=2, bias=False)`, ogʻirliklar 1, kirish `ones(1,1,8,8)`; $(k, s) = (3, 2), (4, 2), (2, 2)$ uchun chiqish shakli va `z[6:10, 6:10]` blokini chop eting.
22. 128×128 maydonda tomoni $a \in \{5, 10, 20, 50\}$ kvadrat (`T[30:30+a, 30:30+a]`) `np.roll` bilan $d \in \{1, 2\}$ gorizonta va $(1, 1)$ diagonal siljitsin. IoU ni $(a - d)/(a + d)$ bilan solishtiring.
23. `soft_dice_loss(logits, target, C, eps=1.0)` (softmax, one-hot, batch va fazo boʻyicha yigʻindi) yozing. 10-topshiriqni tekshiring (`logits=log p`, `eps=0`); `F.cross_entropy(..., weight=w)`; hamda 95 fon +5 obyekt, hamma joyda $p = (0,99; 0,01)$ holati uchun CE va Dice loss.
24. Maʼlumotni yuqoridagi generator bilan yarating. Uch toʻplam uchun shakllar, sinf ulushlari va $w_c \propto 1/\sqrt{n_c}$ ($\sum w_c = 3$), X ning min/max qiymatini chop eting.
25. **Klassik bazaviy chiziqliq**. Test tasvirlari: `gaussian_filter(img, 1.0)` → `threshold_multiotsu(classes=3)` → `np.digitize`. (a) toʻgʻridan-toʻgʻri; (b) avval yoritish tekisligi $ax + by + c$ ni ayiring; eng kichik kvadratlar, 3 iteratsiya, har gal qoldigʻi 70-persentildan kichik piksellar boʻyicha qayta moslash. Butun test toʻplami confusion matrix i boʻyicha mIoU, IoU, PA.
26. `UNet(C=3, c=8, skip=True)`: 3 bosqich, blok = $2 \times (\text{Conv} 3 \times 3, \text{padding } 1 + \text{ReLU})$, pastga `MaxPool2d(2)`, yuqoriga `ConvTranspose2d` ($k = 2, s = 2$), skip — kanal oʻqi boʻyicha `torch.cat`, boshcha `Conv1x1`. $(c, \text{skip}) = (16, \text{True}), (8, \text{True}), (8, \text{False})$ uchun parametrlar soni va chiqish shakli.
27. **U-Net oʻqitish**. `torch.manual_seed(42)` dan keyin `UNet(c=8)`; Adam, `lr=3e-3`, 20 epoxa, batch 10, har epoxa tartibi `np.random.default_rng(42).permutation` (bitta generator, epoxalar boʻyicha davom etadi), CE loss. 1, 5, 10, 20-epoxada oʻrtacha loss va val mIoU; oxirida test mIoU, IoU, PA.
28. **Ablyatsiya**. 27-topshiriq sozlamasida (a) skip siz; (b) 0,5 CE+0,5 Dice loss. Test mIoU va har sinf IoU jadvali; doira sinfi xatolari $(1 - \text{IoU}_2)$ nisbatini hisoblang. Xulosa yozing.
29. **Post-processing**. 25(b) natijasiga: har sinf uchun `cv2.connectedComponentsWithStats` (4-bogʻlanish) bilan maydoni < 30 boʻlgan komponentlarni fon qiling, soʻng 3×3 MORPH_CLOSE bilan faqat fon piksellerini toʻldiring. mIoU oldin/keyin, 0–4 test tasvirlarida 1-sinf komponentlari soni.
30. **Niqobni oʻlchamlash**. Test niqoblarida 1-sinfni fonga almashtiring (faqat 0 va 2 qoladi). `cv2.resize` bilan 32×32 ga (nearest), soʻng 64×64 ga (a) `INTER_NEAREST`, (b) `INTER_LINEAR` va `np rint`. IoU(fon), IoU(doira) va paydo boʻlgan “soxta” 1-yorliqlar sonini chop eting.

Hujjatlar.

- <https://pytorch.org/docs/stable/generated/torch.nn.Conv2d.html>
- <https://pytorch.org/docs/stable/generated/torch.nn.ConvTranspose2d.html>
- https://pytorch.org/docs/stable/generated/torch.nn.functional.cross_entropy.html
- <https://pytorch.org/docs/stable/notes/randomness.html>
- <https://numpy.org/doc/stable/reference/generated/numpy.bincount.html>

- https://scikit-image.org/docs/stable/api/skimage.filters.html#skimage.filters.threshold_multiotsu
- https://docs.scipy.org/doc/scipy/reference/generated/scipy.ndimage.gaussian_filter.html
- https://docs.opencv.org/4.x/d3/dc0/group__imgproc__shape.html
- https://docs.opencv.org/4.x/da/d54/group__imgproc__transform.html